Question	Answer	Marks	Partial Marks
18(c)	*their* $\frac{10}{x-4}$ – *their* $\frac{12}{x}$ = 1 oe	**M1**	
	$10x - 12x + 48 = x^2 - 4x$	**M2**	Correctly multiplying *their* brackets **and** clearing algebraic fractions e.g. $(x-4)\left(\frac{12}{x}+1\right)=10$ leading to $12-\frac{48}{x}+x-4=10$ and then $12x-48+x^2-4x=10x$ or **M1** for correctly clearing, or correctly collecting into a single fraction, two fractions both with different algebraic denominators e.g. $10x - 12(x - 4) = x(x - 4)$ or $\frac{10x-12(x-4)}{x(x-4)}$ [= 1]
	Leading to $0 = x^2 - 2x - 48$	**A1**	With no errors or omissions seen, dep on M3
18(d)	$(x + 6)(x - 8)$	**M2**	**M1** for $x(x - 8) + 6(x - 8)$ or $x(x + 6) - 8(x + 6)$ or $(x + a)(x + b)$ where $ab = -48$ or $a + b = -2$
	–6, 8	**B1**	
18(e)	1.5 or $1\frac{1}{2}$	**1**	FT $\frac{12}{\textit{their}\ 8}$
19	$\frac{1}{9}$	**2**	**M1** for 9^{-1} or for $\frac{1}{3^2}$ or $\frac{1}{\sqrt[3]{729}}$
20	$6\sqrt{3}$ oe	**4**	**B1** for $\tan 30 = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$ or for $\sin 60 = \frac{\sqrt{3}}{2}$ **and** $\sin 30 = \frac{1}{2}$ **M2** for $\frac{6}{\tan 30}$ or $\frac{6\sin(60)}{\sin(30)}$ or **M1** for $\frac{6}{x} = \tan 30$ or $\frac{\sin 60}{x} = \frac{\sin 30}{6}$
21(a)(i)	2**m** – **n**	**1**	
21(a)(ii)	2**m** + **n**	**1**	

Question	Answer	Marks	Partial Marks
21(b)	$\overrightarrow{CD} = 10\mathbf{m} - 5\mathbf{n}$	**M2**	Allow **M2** for equivalents $\overrightarrow{CD} = -2\mathbf{n} + 10\mathbf{m} - 3\mathbf{n}$ or $\overrightarrow{CD} = -(2\mathbf{m}+\mathbf{n}) + 2\mathbf{m} - \mathbf{n} + 10\mathbf{m} - 3\mathbf{n}$ For **M2**, FT *their* **(a)** e.g. $\overrightarrow{CD} = \textit{their}\ \overrightarrow{CO} + \textit{their}\ \overrightarrow{OA} + 10\mathbf{m} - 3\mathbf{n}$ or **M1** for correct route for $\overrightarrow{CD}$ using the lines of the diagram with $\overrightarrow{AD}$ e.g. $\overrightarrow{CD} = \overrightarrow{CA} + \overrightarrow{AD}$ oe
	$\overrightarrow{CD} = 5\overrightarrow{OA}$ leading to CD is parallel to OA [$\therefore$ $OACD$ is a trapezium]	**A1**	Dependent on **M2**
22(a)	[n =] 3, [q =] – 6	**2**	**B1** for each correct value
22(b)	(1, 4) and (3, 0)	**4**	**B3** for (1, 4) or (3, 0) or for two correct values of x or **M2** for [3](x – 1)(x – 3) [= 0] oe or $x = \dfrac{-(-12) \pm \sqrt{(-12)^2 - 4\times3\times9}}{2\times3}$ oe or **M1** for writing $\dfrac{dy}{dx} = 0$ or for $3x^2 - 12x + 9 = 0$
23	$\dfrac{2x}{x-5}$ final answer	**3**	**B1** for $2x(x + 5)$ **B1** for $(x – 5)(x + 5)$